

第16讲. 重积分计算.

1. 1, 2, 3, 5, 6(1), (2), (3), (5), 7(2), (3), 8(2), (3).

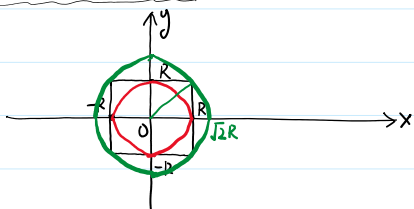
.. = 重积分计算.

例1. 计算 $I = \int_{-\infty}^{+\infty} e^{-x^2} dx$

解: $\because I = \int_{-\infty}^{+\infty} e^{-x^2} dx = \lim_{R \rightarrow +\infty} \int_{-R}^R e^{-x^2} dx.$

$$\begin{aligned} \text{令 } I_R &= \int_{-R}^R e^{-x^2} dx. \quad \text{则 } I_R^2 = \int_{-R}^R e^{-x^2} dx \cdot \int_{-R}^R e^{-y^2} dy = \int_{-R}^R e^{-x^2} \left(\int_{-R}^R e^{-y^2} dy \right) dx \\ &= \iint_D e^{-(x^2+y^2)} dx dy \end{aligned}$$

其中 $D = [-R, R] \times [-R, R].$



$$D_1 = \{(x, y) \mid x^2 + y^2 \leq R^2\}, \quad D_2 = \{(x, y) \mid x^2 + y^2 \leq 2R^2\}.$$

$$D_1 \subset D \subset D_2$$

$$\therefore \iint_{D_1} e^{-(x^2+y^2)} dx dy \leq \iint_D e^{-(x^2+y^2)} dx dy \leq \iint_{D_2} e^{-(x^2+y^2)} dx dy$$

$$\text{令 } \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad \text{则 } D'_1 = \{(r, \theta) \mid 0 \leq \theta < 2\pi, r \in [0, R]\}$$

$$D'_2 = \{(r, \theta) \mid 0 \leq \theta < 2\pi, r \in [0, \sqrt{2}R]\}.$$

$$\therefore \iint_{D_1} e^{-(x^2+y^2)} dx dy = \iint_{D'_1} e^{-r^2} r dr d\theta = \int_0^{2\pi} d\theta \int_0^R e^{-r^2} r dr = 2\pi \left(-\frac{1}{2} e^{-r^2} \Big|_0^R \right) = \pi(1 - e^{-R^2})$$

$\downarrow$
 π

$$\iint_{D_2} e^{-(x^2+y^2)} dx dy = \iint_{D'_2} e^{-r^2} r dr d\theta = \int_0^{2\pi} d\theta \int_0^{\sqrt{2}R} e^{-r^2} r dr = \pi(1 - e^{-2R^2}) \rightarrow \pi$$

$(R \rightarrow +\infty)$

$$\therefore \lim_{R \rightarrow +\infty} I_R^2 = \pi$$

$$\therefore \lim_{R \rightarrow +\infty} I_R = \sqrt{\pi}.$$

$$\therefore \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}.$$

$$\therefore \int_0^{+\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

$$\int_0^{+\infty} e^{-x^2} dx + \int_{-\infty}^0 e^{-x^2} dx$$

例2. 在什么样的区域 $D \subset \mathbb{R}^2$ 上, $\iint_D (4 - x^2 - 2y^2) dx dy$ 值最大?

例2. 求什么样的 α , 在区域 $D \subset \mathbb{R}^2$ 上, $\iint_D (4-x-2y) dx dy$ 值最大?

解: $4 - x^2 - 2y^2 \geq 0$.

$$D = \{(x, y) \mid x^2 + 2y^2 \leq 4\}.$$

$$\begin{aligned} I &= \iint_D (4 - x^2 - 2y^2) dx dy \\ &= 2\sqrt{2} \int_0^{2\sqrt{2}} d\theta \int_0^1 (4 - 4r^2) r dr \\ &= 4\sqrt{2}\pi \int_0^1 (4r - 4r^3) dr \\ &= 4\sqrt{2}\pi (2r^2 - r^4) \Big|_0^1 = 4\sqrt{2}\pi. \end{aligned}$$

$\therefore$ 三重积分.

$$\Omega, \quad T = \{\Delta\Omega_1, \dots, \Delta\Omega_n\}.$$

$$\|T\| = \max \{ \text{diam} \Delta\Omega_i \mid i=1, 2, \dots, n \}$$

$$\forall (x_i, y_i, z_i) \in \Delta\Omega_i.$$

ΔV_i — $\Delta\Omega_i$ 体积.

$$M = \lim_{\|T\| \rightarrow 0} \sum_{i=1}^n \rho(x_i, y_i, z_i) \Delta V_i$$

定义1. 设 $f(x, y, z)$ 定义在有界闭区域 $\Omega \subset \mathbb{R}^3$ 上. 任意分割 Ω . $T = \{\Delta\Omega_1, \dots, \Delta\Omega_n\}$.

$$\Delta V_i \text{ 表示 } \Delta\Omega_i \text{ 体积,} \quad \|T\| = \max \{ \text{diam} \Delta\Omega_i \mid i=1, 2, \dots, n \}.$$

$$\forall (x_i, y_i, z_i) \in \Delta\Omega_i.$$

若 $\lim_{\|T\| \rightarrow 0} \sum_{i=1}^n f(x_i, y_i, z_i) \Delta V_i$ 存在, 且与划分方式 T 无关. 与点选取无关.

2) $f(x, y, z)$ 在 Ω 上可积. 记为 $\iiint_{\Omega} f(x, y, z) dv$. 简记为 $\iiint_{\Omega} f dv$.

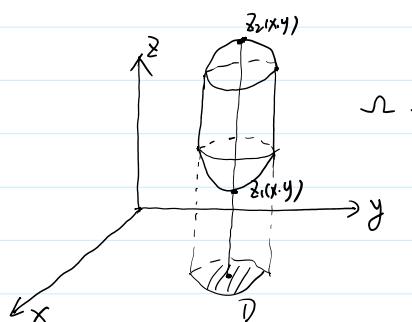
dv — 体积微元.

$$dv = dx dy dz.$$

$$\iiint_{\Omega} f(x, y, z) dv = \iiint_{\Omega} f(x, y, z) dx dy dz.$$

2. 计算.

①. 先-后=法
 $\downarrow$ 定积分
 $\downarrow$ 二重积分



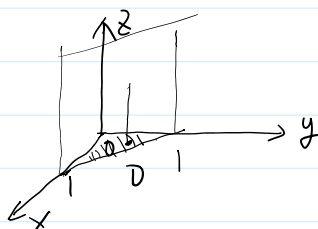
$$\forall (x,y) \in D$$

$$\Omega = \{ (x,y,z) \mid \forall (x,y) \in D, z_1(x,y) \leq z \leq z_2(x,y) \}$$

$$\iiint_{\Omega} f(x,y,z) dx dy dz = \iint_D dx dy \int_{z_1(x,y)}^{z_2(x,y)} f(x,y,z) dz$$

例3. 计算 $I = \iiint_{\Omega} xy dx dy dz$

其中 Ω : $z=xy$, $x+y=1$, $z=0$ 围成. ($x>0, y>0$)



$$\Omega = \{ (x,y,z) \mid \forall (x,y) \in D, 0 \leq z \leq xy \}$$

$$\therefore I = \iiint_{\Omega} xy dx dy dz = \iint_D dx dy \int_0^{xy} xy dz$$

$$= \iint_D x^2 y^2 dx dy = \int_0^1 dx \int_0^{1-x} x^2 y^2 dy$$

$$= \int_0^1 x^2 \frac{1}{3} (1-x)^3 dx = \frac{1}{180}$$

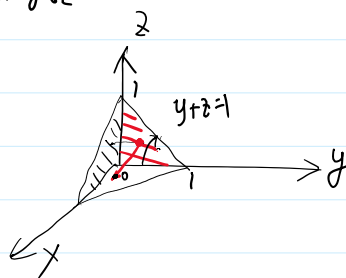
例4. 计算 $I = \int_0^1 dx \int_0^{1-x} dz \int_0^{1-x-z} (1-y) e^{-(1-y-z)^2} dy$

解:

$$\Omega = \{ (x,y,z) \mid x+y+z \leq 1, x>0, y>0, z>0 \}$$

$$I = \iiint_{\Omega} (1-y) e^{-(1-y-z)^2} dx dy dz$$

$$I = \iiint_{\Omega} \underbrace{(1-y)e^{-(1-y-z)^2}}_{\text{被积函数}} dx dy dz$$



$$\Omega = \{ (x, y, z) \mid \forall (y, z) \in D, \quad 0 \leq x \leq 1-y-z \}$$

$$I = \iint_D dy dz \int_0^{1-y-z} \underbrace{(1-y)e^{-(1-y-z)^2}}_{\text{被积函数}} dx$$

$$= \iint_D \underbrace{(1-y)e^{-(1-y-z)^2}}_{\text{被积函数}} \underbrace{(1-y-z)}_{\text{积分上限}} dy dz$$

$$= \int_0^1 (1-y) dy \int_0^{1-y} \underbrace{(1-y-z)}_{\text{被积函数}} \underbrace{e^{-(1-y-z)^2}}_{\text{被积函数}} dz$$

$$= \int_0^1 (1-y) \left(-\frac{1}{2} \right) \left(e^{-(1-y-z)^2} \Big|_{z=0}^{z=1-y} \right) dy$$

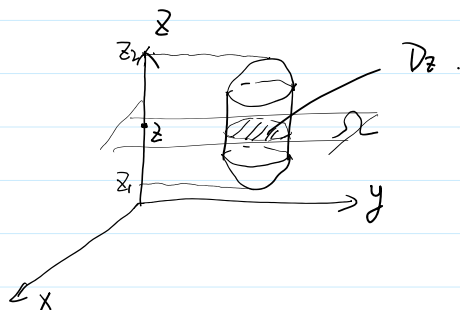
$$= -\frac{1}{2} \int_0^1 (1-y) (1 - e^{-(1-y)^2}) dy$$

$$= \frac{1}{2} \int_0^1 \underbrace{(1-y)e^{-(1-y)^2}}_{\text{被积函数}} dy - \frac{1}{2} \int_0^1 (1-y) dy$$

$$= \frac{1}{4e}$$

(2). “先=后”法.

↓
定积分



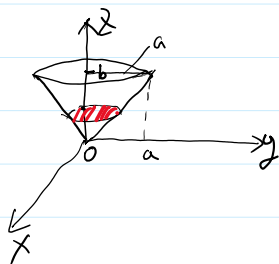
$$\Omega = \{ (x, y, z) \mid z_1 \leq z \leq z_2, \quad (x, y) \in D_z \}$$

$$I = \int_{z_1}^{z_2} dz \iint_{D_z} f(x, y, z) dx dy$$

例5. 计算 $I = \iiint_{\Omega} z \, dx \, dy \, dz$

$\Omega: a^2 z^2 = b^2 (x^2 + y^2), \quad z = b$ 围成锥体. ($a > 0, b > 0$)

解:



$$\Omega = \{ (x, y, z) \mid 0 \leq z \leq b, \quad \forall (x, y) \in D_z \}$$

$$D_z = \{ (x, y) \mid x^2 + y^2 \leq \frac{a^2}{b^2} z^2 \}$$

$$I = \int_0^b z \, dz \iint_{D_z} dx \, dy = \pi \int_0^b z \cdot \frac{a^2}{b^2} z^2 \, dz = \frac{a^2}{b^2} \pi \cdot \frac{1}{4} b^4 = \frac{\pi a^2 b^2}{4}$$

3. 坐标变换.

定理1. 设 $D, \Omega \subset \mathbb{R}^3$ 有界闭区域.

连续可微双射 $\varphi: \begin{cases} x = x(u, v, w) \\ y = y(u, v, w) \\ z = z(u, v, w) \end{cases} \quad (u, v, w) \in \Omega \longrightarrow D$

且 $\forall (u, v, w) \in \Omega, \quad \frac{\partial(x, y, z)}{\partial(u, v, w)} \Big|_{(u, v, w)} \neq 0$.

若 $f(x, y, z) \in C(D)$. (2)

$$\iiint_D f(x, y, z) \, dx \, dy \, dz = \iiint_{\Omega} f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du \, dv \, dw$$

例6. 计算 $I = \iiint_{\Omega} \cos(ax + by + cz) \, dx \, dy \, dz$.

$$\Omega = \{ (x, y, z) \mid x^2 + y^2 + z^2 \leq 1 \}$$

解: $u = \frac{1}{\lambda}(ax + by + cz) = \frac{1}{\lambda}(a, b, c) \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \sqrt{a^2 + b^2 + c^2} = \lambda$

$$V = \begin{pmatrix} \frac{a}{\lambda} & \frac{b}{\lambda} & \frac{c}{\lambda} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix}$$

$$V \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} u \\ v \\ w \end{pmatrix}$$

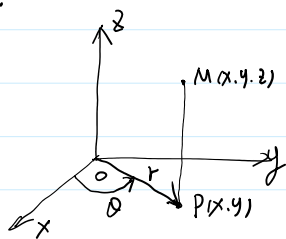
$$UU^T = U^T U = I.$$

$$u^2 + v^2 + w^2 = (u, v, w) \begin{pmatrix} u \\ v \\ w \end{pmatrix} = (x, y, z) U^T U \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x^2 + y^2 + z^2 \leq 1$$

$$\therefore I = \iiint_{u^2+v^2+w^2 \leq 1} \cos(\lambda u) du dv dw = \int_{-1}^1 \cos(\lambda u) du \iint_{v^2+w^2 \leq 1-u^2} dv dw$$

$$= 2 \int_{-1}^1 \cos(\lambda u) (1-u^2) du = \frac{4\pi}{\lambda^2} \left(\frac{\sin \lambda}{\lambda} - \cos \lambda \right).$$

1°. 柱坐标变换.



$$\|\vec{OP}\| = r.$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$$

$$M(r, \theta, z)$$

$$\left| \frac{D(x, y, z)}{D(r, \theta, z)} \right| = r$$

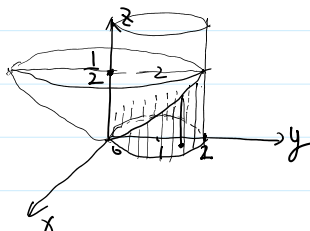
$$\iiint_{\Omega} f(x, y, z) dx dy dz = \iiint_{\Omega'} f(r \cos \theta, r \sin \theta, z) r dr d\theta dz$$

$$\Omega' : (r, \theta, z)$$

例7. 计算 $I = \iiint_{\Omega} \sqrt{x^2 + y^2} dx dy dz$.

$$\Omega : x^2 + (y-1)^2 \leq 1, \quad 8z = x^2 + y^2. \quad z \Rightarrow \text{圆柱面}.$$

解:



$$\begin{aligned} x^2 + y^2 &\leq 2y \\ r^2 &\leq 2r \sin \theta \end{aligned}$$

$$D = \{(x, y) \mid x^2 + (y-1)^2 \leq 1\}.$$

$$\Omega = \left\{ (x, y, z) \mid \forall (x, y) \in D, \quad 0 \leq z \leq \frac{x^2 + y^2}{8} \right\}.$$

$$I = \iint_D \sqrt{x^2 + y^2} dx dy \int_0^{\frac{x^2 + y^2}{8}} dz$$

$$= \iint_D \sqrt{x^2 + y^2} \cdot \frac{x^2 + y^2}{8} dx dy$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$= \iint_D \sqrt{x^2+y^2} \cdot \frac{x^2+y^2}{8} dx dy$$

$$= \int_0^{2\pi} d\theta \int_0^{2\sin\theta} \frac{r^3}{8} r dr = \frac{64}{75}$$

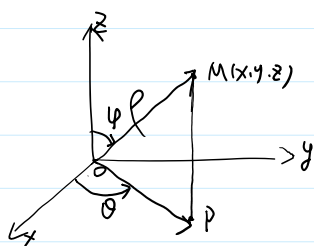
$$\begin{cases} y = r \sin \theta \end{cases}$$

$$D' = \{(r, \theta) \mid \theta \in [0, 2\pi], r \in [0, 2\sin\theta]\}$$

$$\Omega' = \{(r, \theta, z) \mid \theta \in [0, 2\pi], r \in [0, 2\sin\theta], 0 \leq z \leq \frac{r^2}{8}\}$$

$$I = \int_0^{2\pi} d\theta \int_0^{2\sin\theta} r dr \int_0^{\frac{r^2}{8}} r dz$$

2°. 球坐标系,



$$\|\vec{OM}\| = \rho$$

$$\|\vec{OP}\| = \rho \sin \varphi$$

$$\begin{cases} \varphi \in [0, \pi] \\ \theta \in [0, 2\pi] \\ \rho \geq 0 \end{cases}$$

$$\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \varphi \end{cases}$$

$$M(\rho, \varphi, \theta)$$

$$x^2 + y^2 + z^2 = \rho^2$$

$$\frac{D(x, y, z)}{D(\rho, \varphi, \theta)} = \begin{vmatrix} \sin \varphi \cos \theta & \rho \cos \varphi \cos \theta & -\rho \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & \rho \cos \varphi \sin \theta & \rho \sin \varphi \cos \theta \\ \cos \varphi & -\rho \sin \varphi & 0 \end{vmatrix}$$

$$= \rho^3 \sin^3 \varphi \sin^2 \theta + \rho^3 \sin \varphi \cos^2 \varphi \cos^2 \theta + \rho^3 \sin \varphi \cos^2 \varphi \sin^2 \theta + \rho^3 \sin \varphi \cos^2 \varphi$$

$$= \rho^3 \sin^3 \varphi + \rho^3 \sin \varphi \cos^2 \varphi = \rho^3 \sin \varphi$$

$$dx dy dz = \rho^2 \sin \varphi d\rho d\varphi d\theta$$

例8. 求 $I = \iiint_{\Omega} (x + 3y + 5z) dx dy dz$

$$\Omega = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq R^2, x \geq 0, y \geq 0, z \geq 0\}$$

解: $I = \iiint_{\Omega} x dx dy dz + 3 \iiint_{\Omega} y dx dy dz + 5 \iiint_{\Omega} z dx dy dz$

$$= 9 \iiint_{\Omega} x dx dy dz = 9 \iiint_{\Omega} z dx dy dz$$

$$= \underbrace{9 \iiint_{\Omega} x \, dx \, dy \, dz} = \underbrace{9 \iiint_{\Omega} z \, dx \, dy \, dz}.$$

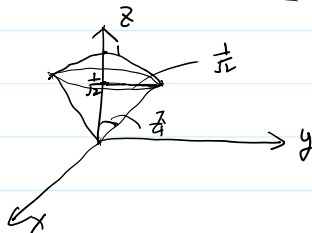
$$\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \varphi \end{cases} \quad \Omega' = \left\{ (\rho, \varphi, \theta) \mid 0 \leq \theta \leq \frac{\pi}{2}, \varphi \in [0, \frac{\pi}{2}], \rho \in [0, R] \right\}$$

$$\begin{aligned} \therefore I &= 9 \iiint_{\Omega} z \, dx \, dy \, dz = 9 \int_0^{\frac{\pi}{2}} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^R \rho \cos \varphi \cdot \rho^2 \sin \varphi \, d\rho \\ &= \frac{9\pi R^4}{16}. \end{aligned}$$

例 9. 计算 $I = \iiint_{\Omega} (x + y^2 + z) \, dx \, dy \, dz$

$\Omega: \underbrace{z = \sqrt{x^2 + y^2}}_{\text{圆锥}}, \underbrace{z = \sqrt{1 - x^2 - y^2}}_{\text{半球}}$ 围成.

解:



$$x^2 + y^2 = 1 - x^2 - y^2$$

$$x^2 + y^2 = \frac{1}{2}$$

$$\begin{aligned} I &= \iiint_{\Omega} x \, dx \, dy \, dz + \iiint_{\Omega} y^2 \, dx \, dy \, dz + \iiint_{\Omega} z \, dx \, dy \, dz \\ &= \iiint_{\Omega} z \, dx \, dy \, dz = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^1 \rho \cos \varphi \cdot \rho^2 \sin \varphi \, d\rho = \frac{\pi}{8}. \end{aligned}$$

$$\Omega' = \left\{ (\rho, \varphi, \theta) \mid 0 \leq \theta \leq 2\pi, \varphi \in [0, \frac{\pi}{4}], \rho \in [0, 1] \right\}$$

$$\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \varphi \end{cases}$$

例 10. 求曲面 $(x^2 + y^2 + z^2)^3 = a^3 xyz$ ($a > 0$) 所围立体体积.

解: $V = \iiint_{\Omega} dx \, dy \, dz = 4 \iiint_{\Omega'} dx \, dy \, dz = 4 \int_0^{\frac{\pi}{2}} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^{a \sqrt[3]{\sin^2 \varphi \cos \varphi \sin \theta \cos \theta}} \rho^2 \sin \varphi \, d\rho = \frac{a^3}{6}.$

Ω_1 第 I 卦限区域.

$$\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \varphi \end{cases} \quad \rho = a \sqrt[3]{\sin^2 \varphi \cos \varphi \sin \theta \cos \theta}$$

$$\Omega' = \left\{ (\rho, \varphi, \theta) \mid 0 \leq \theta \leq \frac{\pi}{2}, \varphi \in [0, \frac{\pi}{2}], \rho \in [0, a \sqrt[3]{\sin^2 \varphi \cos \varphi \sin \theta \cos \theta}] \right\}$$

$$(x^2 + y^2 + z^2)^2 = a^2(x^2 + y^2 - z^2)$$

$$V = \iiint_{\Omega} dx dy dz$$

$$= 8 \iiint_{\Omega_1} dx dy dz$$

$$\rho^2 = a^2 \cdot (-\cos 2\varphi) \geq 0$$

$$\cos 2\varphi \leq 0.$$

$$\frac{\pi}{2} \leq 2\varphi \leq \frac{3\pi}{2}$$

$$\frac{\pi}{4} \leq \varphi \leq \frac{3\pi}{4}$$